



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PRANAV AGENCY UNIT II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

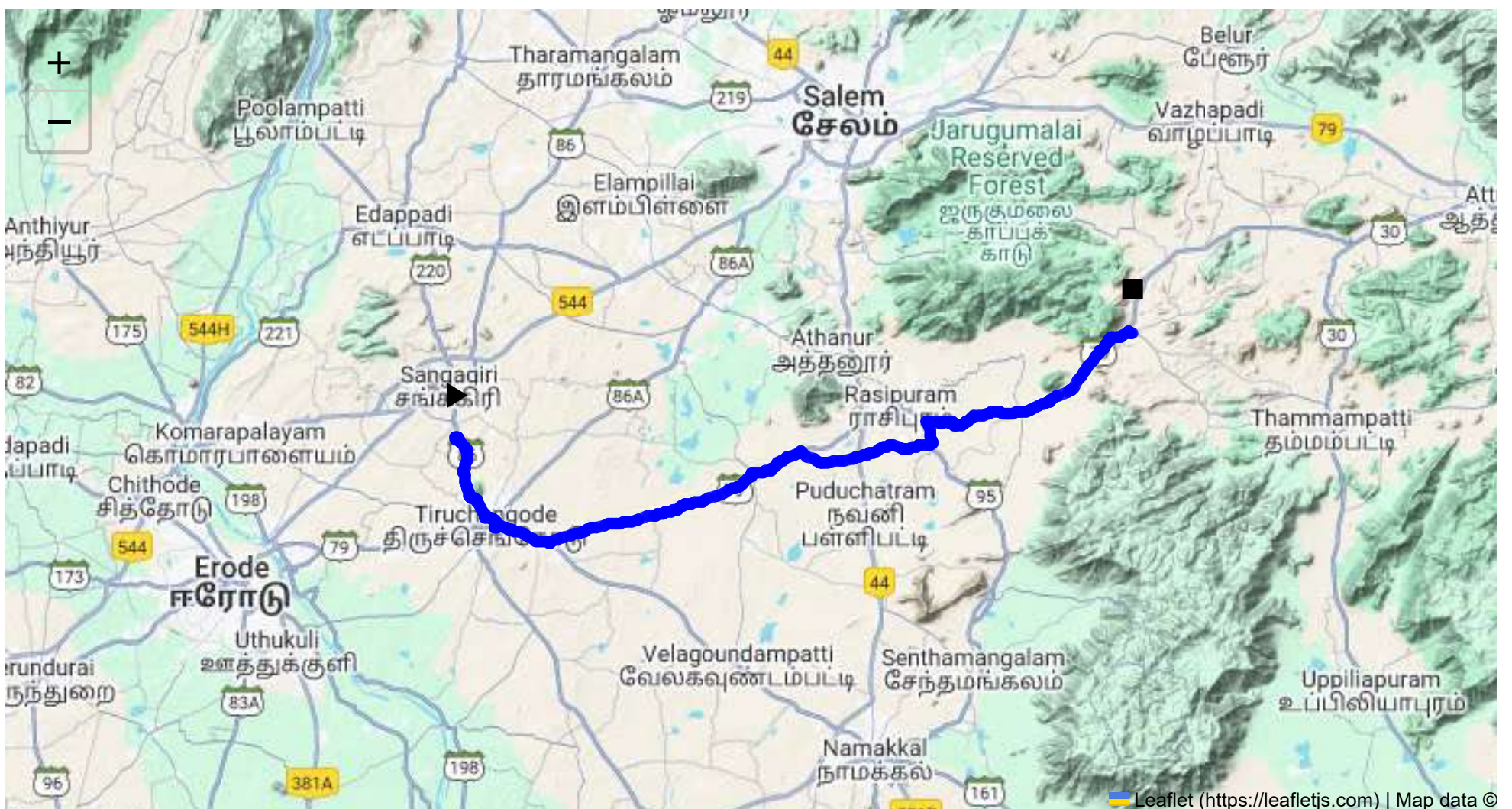
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 65.18 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (11.5104058, 78.34693408)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

This route begins at CVQF+23W, Sangagiri, and proceeds northeast towards G86W+4RC, Mettala Rd, passing via 20A, Old Salem Rd, MUTHUMANHOPPU, and MUTHUSELVAM NAGAR, in Tiruchengode. The journey covers around 65.18 kilometers through a mixture of urban and semi-rural environments. This

area has a mix of two-lane roads and smaller local roads, which can impact vehicle movement, especially for heavy vehicles.

2. Typical Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate with high temperatures and seasonal monsoons. The primary weather-related hazards include:

- **Monsoon Season (June to September):** Heavy rain can lead to flooding and reduced visibility.
- **Cyclones and Storms:** Can occasionally affect the region during the late monsoon and post-monsoon period (October to December).

3. Traffic Patterns

- **Peak Traffic Hours:** Typically from 8:00 AM to 10:00 AM and 5:00 PM to 8:00 PM.
- **Congestion-Prone Areas:**
 - Urban areas around Tiruchengode.
 - Entry and exit points to larger roads from Old Salem Rd and smaller local roads may experience bottlenecks.

4. Road Quality and Infrastructure

- **20A and Old Salem Rd:** Generally maintained but may have sections with potholes and uneven surfaces, especially post-rains.
- **Local Roads:** May have limited signage and narrower stretches, requiring cautious navigation.

5. Alternative Routes for Emergencies

- Alternative routes include bypasses through Erode for a longer but potentially quicker route during obstructions.
- Use GPS navigation apps which provide real-time updates on traffic and road conditions to guide the rerouting.

6. Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has specific regulations for hazardous materials:

- Restrictions on timings for transport through urban areas.
- Requirement for mandatory permits and adherence to signage.

7. Historical Incidents

While there are no recent major incidents involving hazardous materials on this specific route, nearby urban areas have witnessed accidents due to heavy traffic. Drivers should maintain caution, especially in congested zones.

8. Environmental Considerations and Sensitive Areas

- Areas near water bodies are sensitive and must be navigated with caution to prevent contamination in case of spills.
- Agricultural land is prevalent, where spillage could harm crops.

9. Communication Coverage

- Most urban and semi-urban areas have good cellular coverage, but occasional dead zones might be encountered in rural sections or near dense foliage between Tiruchengode and Sangagiri.

10. Estimated Emergency Response Times

- Urban areas: 15–30 minutes.
- Semi-rural portions: 30–45 minutes.
- In remote or less accessible areas, times might increase beyond 45 minutes.

11. Overall Summary of Risk Assessment

This route presents moderate risks for transporting hazardous materials, largely due to:

- Possible weather-related disruptions.
- Congestion in urban areas.
- Mixed road quality requiring vigilance.

Recommendations:

- Plan travel times to avoid peak hours.
- Monitor weather forecasts and road condition alerts.
- Ensure compliance with all regulatory requirements for safety.
- Optimize communication strategies to mitigate potential dead zone issues.

By adhering to these guidelines and remaining alert, truck drivers can safely navigate this route while minimizing potential hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	Medium	11.38346, 77.89975	30 KM/Hr
5	Turn	Medium	11.38302, 77.89995	30 KM/Hr
6	Turn	High	11.38188, 77.90257	15 KM/Hr
7	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
8	Turn	Medium	11.36745, 77.93951	30 KM/Hr
9	Turn	Medium	11.41332, 78.08072	30 KM/Hr
10	Turn	High	11.41463, 78.08067	15 KM/Hr
11	Turn	High	11.42964, 78.11587	15 KM/Hr
12	Turn	Medium	11.42454, 78.15291	30 KM/Hr
13	Turn	High	11.43542, 78.20754	15 KM/Hr
14	Blind Spot	Blind Spot	11.45100, 78.20385	10 KM/Hr
15	Turn	Medium	11.44925, 78.22288	30 KM/Hr
16	Turn	Medium	11.44813, 78.22305	30 KM/Hr
17	Turn	Medium	11.44793, 78.22320	30 KM/Hr
18	Turn	Medium	11.49628, 78.32342	30 KM/Hr
19	Turn	Medium	11.50497, 78.33079	30 KM/Hr
20	Turn	Medium	11.50769, 78.33291	30 KM/Hr
21	Turn	Medium	11.50860, 78.33951	30 KM/Hr
22	Turn	High	11.51180, 78.34422	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
10	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38302, 77.89995



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38188, 77.90257



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.36745, 77.93951



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41332, 78.08072



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41463, 78.08067



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42454, 78.15291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43542, 78.20754



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45100, 78.20385



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44925, 78.22288



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44813, 78.22305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44793, 78.22320



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49628, 78.32342



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50497, 78.33079



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50769, 78.33291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50860, 78.33951



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51180, 78.34422
